PH.40121 **FALL 2026**

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Lecture 3: Classification of Admissible Field Equations

Munhyeok Kang 2026-09-14–2026-09-20

Overview

Second-order, rotationally invariant field operators are reduced by staticity, flat-slice, and weak-field constraints.

1 Physical assumptions

All coordinate conditions and boundary conditions are stated before a parameter is fitted.

$$K = \sum_{n=1}^{75} c_n k_n$$

2 Field equations

The first two displayed relations establish the notation used in the derivation.

$$K_{28}[\Phi] = \kappa\rho$$

3 Observable limit

Weak-field, static, and point-source limits are checked independently.

$$\Delta\chi^2 = \sum_a (R_a^{new} - R_a^{old})^2$$

4 Parameter dependence

Parameter combinations are compared through derivatives of the reference observables.

5 Example

The calculation is repeated for one analytic case and one tabulated data point.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Einstein–Mythos paper D, §§1–6
- Candidate Reference 47, historical facsimile
- SGT9 appendix K